

Gezheng XU

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Highlights

Research Profile: Trustworthy machine learning under distribution shifts, with a focus on domain adaptation, robustness, fairness, uncertainty-aware learning, and foundation models.

Current Research Interests: Trustworthy foundation models and agentic AI systems, with a focus on safety under distribution shifts, uncertainty-aware evaluation and monitoring, robust adaptation, privacy-preserving learning, and reliable deployment in scientific and high-stakes domains.

Selected Recognition: ICML 2026 Spotlight, ICLR 2023 Spotlight, Fields Postdoctoral Fellowship Award, Robert and Ruth Lumsden Graduate Award

Education

University of Western Ontario , London, Canada <i>Ph.D. in Computer Science</i> Advisor: Prof. Boyu Wang and Prof. Charles Ling.	Sept. 2021 – Feb. 2026 Grade: 94.3/100
Beihang University , Beijing, China <i>Master in Industrial Engineering, Sino-French Engineering School</i> Advisor: Prof. Wenge Rong.	Sept. 2018 – Jan. 2021 Grade: 91.3/100
Paris-Saclay University (École Centrale Paris) , Paris Area, France <i>Engineer's Degree (Diplôme d'Ingénieur, Double Degree)</i>	Sept. 2017 – Jul. 2019 GPA: 4.07/4.33 (A)
Beihang University , Beijing, China <i>Bachelor of Science in Mathematics and Applied Mathematics</i> <i>Sino-French Engineering School</i>	Sept. 2014 – Jun. 2018 GPA: 3.73/4.00 Rank 4/111

Honors & Awards

- Fields Postdoctoral Fellowship Award, Western University, 2026 CAD 25,000
- Robert and Ruth Lumsden Graduate Award, Western University, 2024–2025
- Graduate Student Teaching Award, Western University, 2023–2024
- Excellent Graduate Student, Beihang University, 2018
- China Scholarship Council Scholarship for Outstanding Undergraduate Students, 2017 RMB 100,000
- National Undergraduate Scholarship, 2015–2016 Top 1%
- First-class Academic Scholarship, Beihang University, 2015–2016

Publications

* denotes equal contribution, ⁺ denotes corresponding author(s). Publication tags indicate research themes and data modalities.

Conference Publications

- [C1] **Gezheng Xu**, Qi Chen, Qiuhaio Zeng, Charles Ling, Boyu Wang. Discretized Density-Guided Source-Free Domain Adaptation for Regression. *International Conference on Machine Learning (ICML)*, 2026. (**Spotlight, acceptance rate $\approx 2.2\%$**).[Adaptation] [Robustness] [CV]
- [C2] Jiale Cai, **Gezheng Xu**⁺, Zhihao Li, Ruiyi Fang, Ruizhi Pu, Di Wu, Qicheng Lao, Charles Ling, Boyu Wang⁺. FUSE: Full-spectrum Unlearnable Examples via Spectral Equalization. *International Conference on Machine Learning (ICML)*, 2026. [Privacy/Security] [CV]
- [C3] Linxiao Yang, Xue Jiang, **Gezheng Xu**, Tian Zhou, Min Yang, ZhaoYang Zhu, Linyuan Geng, Zhipeng Zeng, Qiming Chen, Xinyue Gu, Rong Jin, Liang Sun. Baguan-TS: A Sequence-Native In-Context Learning Model

for Time Series Forecasting with Covariates. *International Conference on Machine Learning (ICML)*, 2026. [\[Paper\]](#) [\[Foundation Models\]](#) [\[Structured Data\]](#)

- [C4] Qihao Zeng, Jerry Huang, Peng Lu, Ruiyi Fang, **Gezheng Xu**, Zihao Jing, Yufei Cui, Charles Ling, Gang Niu, Boyu Wang. Attention with Routed-Memory for Learnable Sparse Control. *International Conference on Machine Learning (ICML)*, 2026. [\[Foundation Models\]](#)
- [C5] Zhihao Li, **Gezheng Xu**⁺, Jiale Cai, Ruiyi Fang, Di Wu, Qicheng Lao, Charles Ling, Boyu Wang. When Priors Backfire: On the Vulnerability of Unlearnable Examples to Pretraining. *International Conference on Learning Representations (ICLR)*, 2026. [\[Paper\]](#) [\[Privacy/Security\]](#) [\[Foundation Models\]](#) [\[CV\]](#)
- [C6] Zhihao Li, Jiale Cai, **Gezheng Xu**⁺, Hao Zheng, Qiuyue Li, Fan Zhou, Shichun Yang, Charles Ling, Boyu Wang⁺. Versatile Transferable Unlearnable Example Generator. *Neural Information Processing Systems (NeurIPS)*, 2025. [\[Paper\]](#) [\[Privacy/Security\]](#) [\[CV\]](#)
- [C7] Ruiyi Fang, Bingheng Li, Jingyu Zhao, Ruizhi Pu, Qihao Zeng, **Gezheng Xu**, Charles Ling, and Boyu Wang. Homophily Enhanced Graph Domain Adaptation. *International Conference on Machine Learning (ICML)*, 2025. [\[Paper\]](#) [\[Adaptation\]](#) [\[Structured Data\]](#)
- [C8] **Gezheng Xu**^{*}, Hui Guo^{*}, Li Yi, Charles Ling, Boyu Wang, and Grace Yi. Revisiting Source-Free Domain Adaptation: a New Perspective via Uncertainty Control. *International Conference on Learning Representations (ICLR)*, 2025. [\[Paper\]](#) [\[Code\]](#) [\[Adaptation\]](#) [\[Robustness\]](#) [\[CV\]](#)
- [C9] Qihao Zeng, Jerry Huang, Peng Lu, **Gezheng Xu**, Boxing Chen, Charles Ling, and Boyu Wang. ZETA: Leveraging Z-order Curves for Efficient Top-k Attention. *International Conference on Learning Representations (ICLR)*, 2025. [\[Paper\]](#) [\[Foundation Models\]](#)
- [C10] Ruizhi Pu, **Gezheng Xu**, Ruiyi Fang, Bingkun Bao, Charles Ling, and Boyu Wang. Leveraging Group Classification with Descending Soft Labeling for Deep Imbalanced Regression. *AAAI Conference on Artificial Intelligence (AAAI)*, 2025. (**Oral, acceptance rate $\approx 4.6\%$**). [\[Paper\]](#) [\[Robustness\]](#)
- [C11] **Gezheng Xu**, Qi Chen, Charles Ling, Boyu Wang, Changjian Shui. Intersectional Unfairness Discovery. *International Conference on Machine Learning (ICML)*, 2024. [\[Paper\]](#) [\[Code\]](#) [\[Fairness\]](#) [\[Structured Data\]](#)
- [C12] Qihao Zeng, Wei Wang, Fan Zhou, **Gezheng Xu**, Ruizhi Pu, Changjian Shui, Christian Gagné, Shichun Yang, Charles Ling, Boyu Wang. Generalizing Across Temporal Domains with Koopman Operators. *AAAI Conference on Artificial Intelligence (AAAI)*, 2024. [\[Paper\]](#) [\[Adaptation\]](#)
- [C13] Li Yi^{*}, **Gezheng Xu**^{*}, Pengcheng Xu, Jiaqi Li, Ruizhi Pu, Charles Ling, A. Ian McLeod, Boyu Wang. When Source-Free Domain Adaptation Meets Learning with Noisy Labels. *International Conference on Learning Representations (ICLR)*, 2023. (**Spotlight, acceptance rate $\approx 5\%$**). [\[Paper\]](#) [\[Code\]](#) [\[Adaptation\]](#) [\[Robustness\]](#) [\[CV\]](#)
- [C14] Changjian Shui^{*}, **Gezheng Xu**^{*}, Qi Chen, Jiaqi Li, Charles Ling, Tal Arbel, Boyu Wang, Christian Gagné. On Learning Fairness and Accuracy on Multiple Subgroups. *Neural Information Processing Systems (NeurIPS)*, 2022. [\[Paper\]](#) [\[Code\]](#) [\[Fairness\]](#) [\[CV\]](#) [\[NLP\]](#)

Journal Publications

- [J1] Pengcheng Xu, Li Yi, **Gezheng Xu**, Xi Chen, Ian McLeod, Charles Ling, and Boyu Wang. Uniform Noise Distribution and Compact Clusters: Unveil the Key to Self-Supervised Learning's Success in Label Noise. *Transactions on Machine Learning Research (TMLR)*, accepted. [\[Paper\]](#) [\[Robustness\]](#) [\[CV\]](#)
- [J2] Ruizhi Pu, Lixing Yu, Shaojie Zhan, **Gezheng Xu**, Fan Zhou, Charles Ling, and Boyu Wang. FedELR: When Federated Learning Meets Learning With Noisy Labels. *Neural Networks*, vol. 187, no. 107275, 2025. (IF: 6.3) [\[Paper\]](#) [\[Robustness\]](#) [\[Privacy/Security\]](#) [\[CV\]](#)
- [J3] **Gezheng Xu**^{*}, Li Yi^{*}, Pengcheng Xu, Jiaqi Li, Ruizhi Pu, A. Ian McLeod, Boyu Wang, Charles Ling. Unraveling the Mysteries of Label Noise in Source-Free Domain Adaptation: Theory and Practice. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, vol. 47, no. 5, pp. 4007 - 4023, 2025. (IF: 18.6) [\[Paper\]](#) [\[Code\]](#) [\[Adaptation\]](#) [\[Robustness\]](#) [\[CV\]](#)
- [J4] Boyu Wang, Jorge Mendez, Changjian Shui, Fan Zhou, Di Wu, **Gezheng Xu**, Christian Gagné, Eric Eaton. Gap Minimization for Knowledge Sharing and Transfer. *Journal of Machine Learning Research (JMLR)*, vol. 24, no. 33, pp. 1-57, 2023. (IF: 4.7) [\[Paper\]](#) [\[Code\]](#) [\[Adaptation\]](#)

- [J5] Jiaqi Li*, Haojia Kong*, **Gezheng Xu**, Changjian Shui, Ruizhi Pu, Zhao Kang, Charles X. Ling, Boyu Wang. Label Shift Conditioned Hybrid Querying for Deep Active Learning. *Knowledge-Based Systems (KBS)*, vol. 274, 2023. (IF: 7.6) [\[Paper\]](#) [\[Robustness\]](#) [\[Adaptation\]](#)
- [J6] Changjian Shui*, Ruizhi Pu*, **Gezheng Xu**, Jun Wen, Fan Zhou, Christian Gagné, Charles X. Ling, Boyu Wang. Towards More General Loss and Setting in Unsupervised Domain Adaptation. *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, vol. 35, no. 10, pp. 10140-10150, 2023. (IF: 10.4) [\[Paper\]](#) [\[Adaptation\]](#)
- [J7] **Gezheng Xu**⁺, Wenge Rong, Yanmeng Wang, Yuanxin Ouyang, Zhang Xiong. External Features Enriched Model for Biomedical Question Answering. *BMC Bioinformatics*. 22(1): 272 (2021). (IF: 2.8) [\[Paper\]](#) [\[Adaptation\]](#) [\[NLP\]](#)

PhD Thesis

Trustworthy AI Under Distribution Shifts: Enhancing Generalization and Fairness in Deep Learning. (2026). Retrieved from Western University Institutional Repository [\[Link\]](#).

Academic Experience

Postdoctoral Research Assistant

University of Western Ontario

London, Canada

Feb. 2026 – Present

- Lead research on trustworthy machine learning under distribution shifts, focusing on uncertainty-aware source-free adaptation in both regression and classification settings.
- Develop methods for robust and reliable learning under limited supervision, data accessibility constraints, and target-domain distribution shifts.
- Mentor junior PhD students on trustworthy AI topics, including unlearnable example generation and time-series domain adaptation.

Mentorship

University of Western Ontario

London, Canada

May 2025 – Present

- Mentored junior PhD students on research projects related to trustworthy AI, including project formulation, experimental design, implementation, and manuscript preparation.
- Provided guidance on topics including time-series domain adaptation, unlearnable examples, and robust learning under distribution shifts.

Teaching Assistant

University of Western Ontario

London, Canada

Sept. 2021 – May 2025

- **Introduction to Artificial Intelligence** (Fall 2021, Fall 2022): graded assignments and exams, held office hours, and served as **Head TA** in Fall 2022 by coordinating the TA team, grading tasks, assignments, and the final exam.
- **Applied Logic for Computer Science** (Winter 2022): delivered tutorials and graded assignments.
- **Artificial Intelligence** (2023–2025): held office hours, graded assignments, and evaluated final projects; served as **Head TA** in 2024–2025 by coordinating the TA team, grading logistics, and routine course support.

Research Assistant

University of Western Ontario

London, Canada

Jan. 2021 – Sept. 2021

- Assisted in research on label-shift conditioned hybrid querying for deep active learning.

Industry and Internship Experience

Damo Academy, Alibaba Group

Research Intern

Remote (from Canada)

Oct. 2025 – Apr. 2026

- Conducted research on large-scale time-series foundation models, contributing to Baguan-TS, a sequence-native in-context learning model for time-series forecasting with covariates accepted by ICML 2026.

Murex

Integration Technical Consultant Intern

Beijing, China

Dec. 2019 – Oct. 2020

- Assisted in bank risk control software interface design and workflow integration.

Commercial Aircraft Corporation of China Ltd

AI Department Intern

Beijing, China

Jul. 2019 – Sept. 2019

- Participated in the Pilot Intelligence Instruction Manual project and investigated information retrieval and deep learning methods for aviation-related data.
- Built a BM25-based information retrieval demo for pilot instruction manuals and preliminary aviation data analysis.

Skills & Professional Service

Programming: Python (PyTorch, TensorFlow, NumPy, scikit-learn, pandas), Java, MATLAB, C, XSL/XSLT.

Languages: Chinese (native), English (professional proficiency), and French (professional proficiency).

Leadership: Vice-president of Students Union of École Centrale de Pékin in Beihang University (2016–2017); Head TA for CS3346/CS9146 and CS4442/CS9542.

Professional Service: Reviewer for ICLR (2024–2026), NeurIPS (2022–2026), ICML (2022–2026), AISTATS (2022–2024), TMLR, and ACM Computing Surveys.

Selected Talks: [Oral Presentation](#) at ICLR 2023, Kigali, Rwanda; invited talk at Vector Institute for Artificial Intelligence, 2025.